# Data introduction: EEG and MRI

Generated 22 October 2026 by `make_data_report.py` from `processed/`. Figures are also saved in `reports/figures/`.

**870**
ADNI MRI subjects
294 AD · 576 CN

**67**
BrainLat EEG subjects
35 AD · 32 CN

**65**
ds004504 EEG subjects
36 AD · 29 CN

## 1. Overview

The project compares Alzheimer's disease (AD) with cognitively normal controls (CN) in two modalities. Structural MRI comes from ADNI as FreeSurfer-derived regional measures (no images are used yet). Resting-state EEG comes from two independent public datasets, BrainLat (Latin America, two sites) and OpenNeuro ds004504 (Greece, one site), processed with the same pipeline. The labels are consistent across all three: **dx = AD** / **CN** , **sex = M** / **F** .

| Dataset | Modality | Data form | Location | Size used | AD | CN |
|---|---|---|---|---|---|---|
| ADNI (UCSF FreeSurfer 7, UCSFFSX7) | Structural MRI, 3T T1 | Tabular FreeSurfer measures (324 per subject) | /project/mlresearch/Runkun/MRI → processed/mri | 82 MB | 294 | 576 |
| BrainLat (Synapse syn51549340) | Resting-state EEG | EEGLAB .set/.fdt, 128-ch BioSemi, 512 Hz | /projectnb/mlresearch/Runkun/brainlat_eeg_full | 8.0 GB | 35 | 32 |
| OpenNeuro ds004504 | Resting-state EEG, eyes closed | BIDS, EEGLAB .set, 19-ch 10–20, 500 Hz | /projectnb/mlresearch/Runkun/ds004504 | 2.2 GB | 36 | 29 |

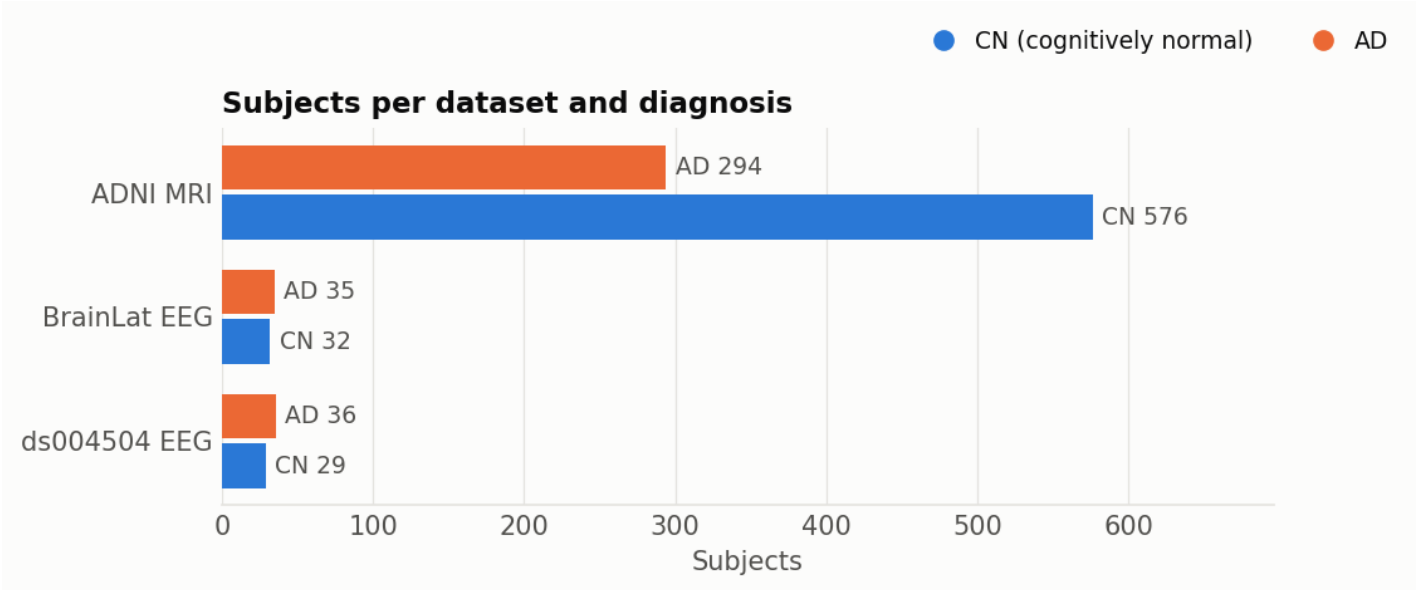



Figure 1. Subjects per dataset. ADNI CN subjects are age- and sex-matched 2:1 to AD.

### How each cohort was selected

| Dataset | Step | AD | CN |
|---|---|---|---|
| BrainLat | Subjects in Records CSVs | 35 | 46 |
| BrainLat | Signal file (.fdt / embedded data) on Synapse | 35 | 32 |
| BrainLat | Loads, harmonized, features extracted | 35 | 32 |
| ds004504 | Participants (Group A / C; 23 FTD excluded) | 36 | 29 |
| ds004504 | Loads, harmonized, features extracted | 36 | 29 |
| ADNI MRI | AD/CN baseline 3T scans, CN age/sex-matched 2:1 | 294 | 576 |

BrainLat: 14 HC recordings are header-only on Synapse (the .fdt signal file does not exist anywhere in the project), so CN drops from 46 to 32. All 35 AD recordings are usable.

## 2. Demographics

| Dataset | Group | n | Age, mean (SD) | Female, n (%) | Education, y | MMSE | p (age) | p (sex) | p (education) |
|---|---|---|---|---|---|---|---|---|---|
| ADNI MRI | CN | 576 | 73.3 (7.8) | 290 (50%) | 16.6 (2.5) | – | | | |
| ADNI MRI | AD | 294 | 73.2 (7.6) | 146 (50%) | 15.5 (2.8) | – | 0.961 | 0.904 | <0.001 |
| BrainLat EEG | CN | 32 | 69.8 (7.7) | 20 (62%) | 15.6 (3.4) | – | | | |
| BrainLat EEG | AD | 35 | 76.5 (7.6) | 22 (63%) | 9.3 (4.6) | – | 0.002 | 1.000 | <0.001 |
| ds004504 EEG | CN | 29 | 67.9 (5.4) | 11 (38%) | not recorded | 30.0 (0.0) | | | |
| ds004504 EEG | AD | 36 | 66.4 (7.9) | 24 (67%) | not recorded | 17.8 (4.5) | 0.383 | 0.039 | – |

p values: Mann–Whitney U for age and education, $\chi^2$ for sex (AD vs CN within each dataset). MMSE is only available for ds004504, where every CN subject scores 30.

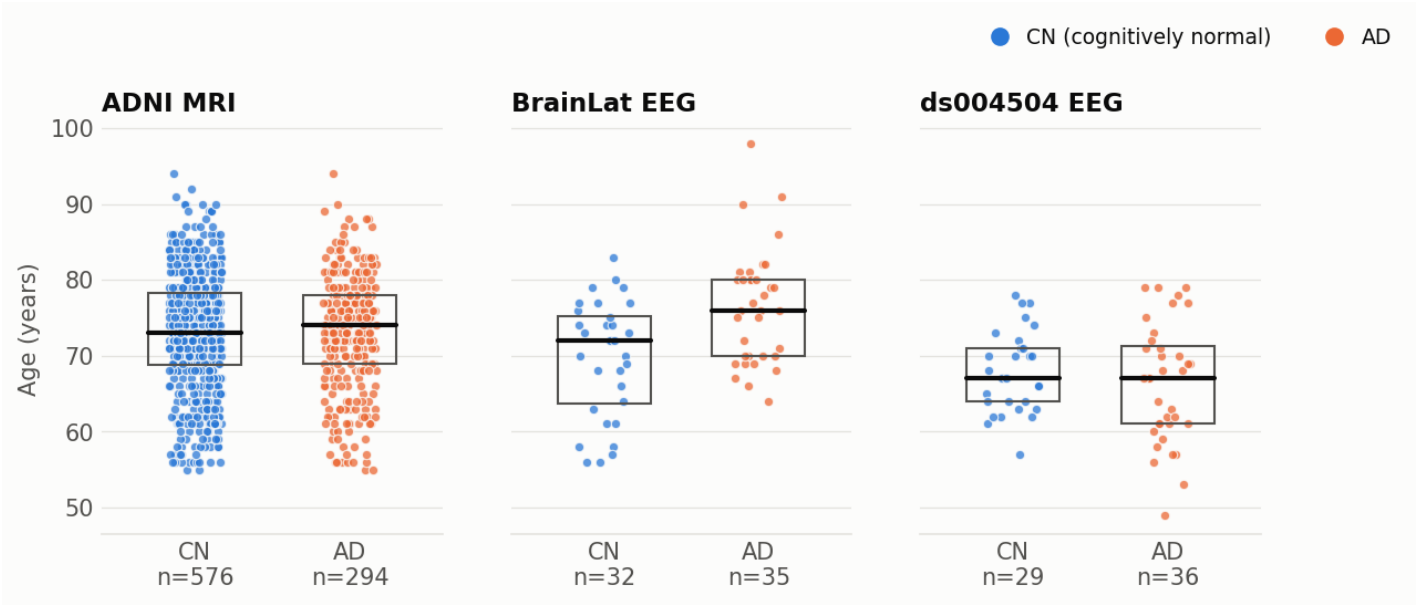



Figure 2. Age by diagnosis. Box = interquartile range, dark bar = median, dots = subjects.

## 3. MRI: ADNI structural measures

Source tables: `All_Subjects_UCSFFSX7` (UCSF FreeSurfer 7 cross-sectional runs), `DXSUM` (diagnosis), `PTDEMOG` (demographics), `DATADIC` and `Key_MRI`. One scan per subject (the earliest that passes QC), 3T only, diagnosis from the closest DXSUM visit (median gap 10 days). Each subject has 324 measures: cortical volume (_CV), surface area (_SA), thickness average (_TA) and SD (_TS), and subcortical volume (_SV).

- All 870 scans are 3T; FreeSurfer 7.2.0 (n=769), 7.4.1 (n=101).
- Only 101 runs carry a UCSF QC rating; the rest are unrated ( `qc_rated = False` ) but kept.
- Scanner make is known for 371 subjects (Siemens 256, GE 65, Philips 49, Other 1).
- Matching worked: age 73.2 vs 73.3 years, ~50% female in both groups. Education still differs (15.5 vs 16.6 years).

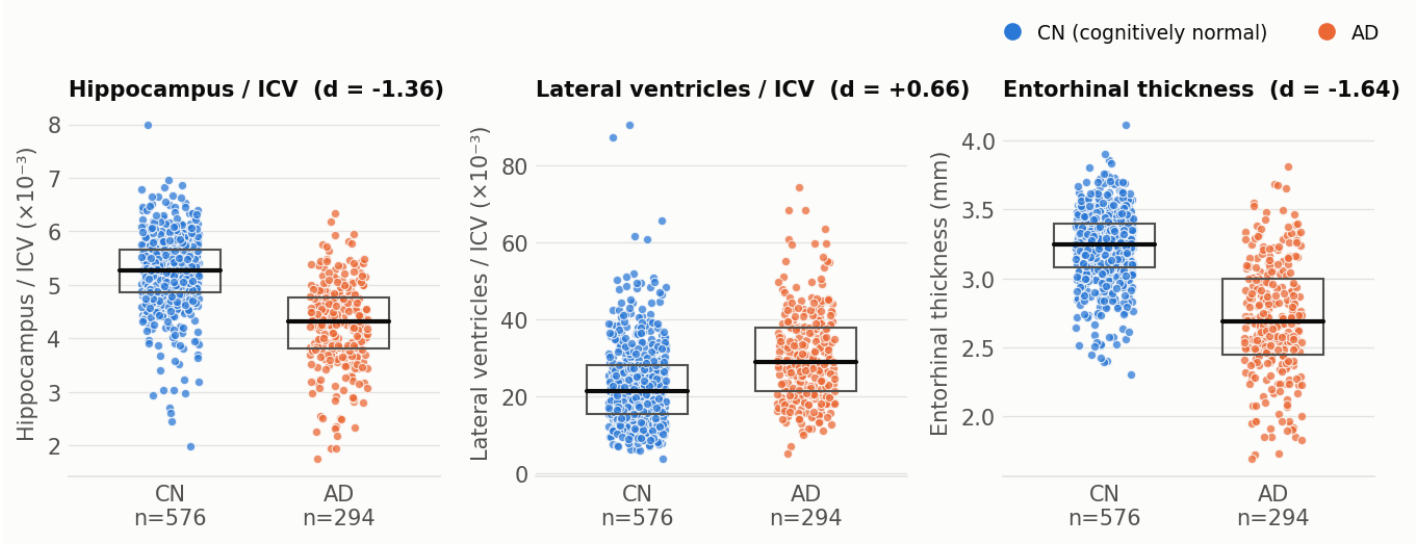



Figure 3. Classic AD atrophy markers: hippocampal and lateral-ventricle volume normalized to intracranial volume (ICV), and mean entorhinal cortical thickness. d = Cohen's d (AD − CN).

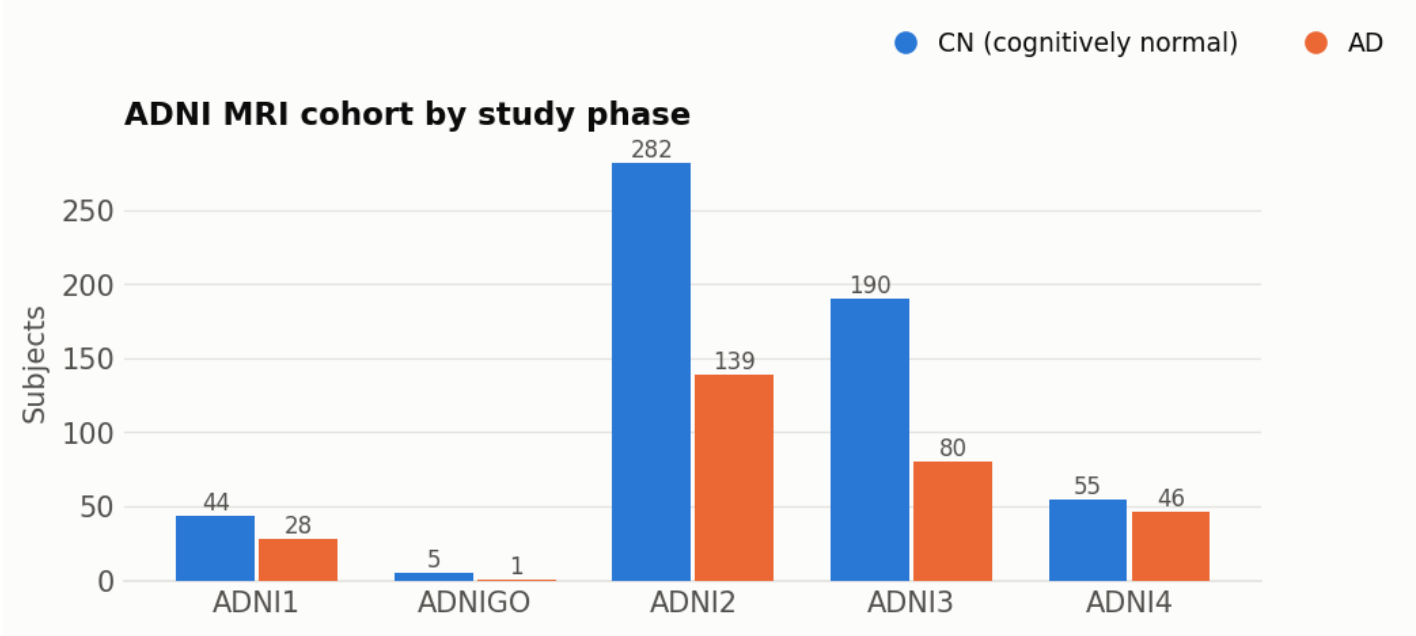



Figure 4. ADNI phase of each subject's scan. The AD:CN ratio varies by phase (CN were matched on age and sex, not phase), so phase/protocol should be a covariate.

## 4. EEG: BrainLat and ds004504

Both EEG datasets run through the same code ( `prepare_brainlat_eeg.py` ; `prepare_ds004504_eeg.py` imports its functions): EEG channels only, average reference, 1–30 Hz band-pass, resample to 256 Hz, first ≤300 s. Features from Welch PSD (2 s windows, 50% overlap): relative power in delta (1–4 Hz), theta (4–8), alpha (8–13) and beta (13–30), globally and per region, theta/alpha ratio, and peak alpha frequency (7–13 Hz).

| | BrainLat | ds004504 |
|---|---|---|
| Recording | AD: `rs-HEP` files · CN: `rs` files | Eyes-closed rest ( `task-eyesclosed` ) |
| Amplifier / montage | BioSemi, 128 channels | Nihon Kohden, 19 channels (10–20; T3/T4/T5/T6 names) |
| Sampling rate | 512 Hz (all) | 500 Hz (all) |
| Sites | AR (Argentina) and CL (Chile) | Single site |
| Raw length, median | AD 424 s · CN 441 s | AD 829 s · CN 827 s |
| Files used | `s6_sub-*_rs[-HEP]_eeg.set` + .fdt | `sub-*/eeg/*_eeg.set` (raw, not derivatives) |

### Channels per scalp region

Regions come from 3-D head coordinates (front > 3 cm, back < –4 cm, lateral > 5 cm = temporal, otherwise central). The two montages differ a lot in density: a BrainLat region averages 19–44 channels, a ds004504 region 1–7. With this rule C3/C4 fall in "temporal", so ds004504 "central" is Cz alone and its central features are noisier than the others.

| | Total | frontal | central | temporal | posterior |
|---|---|---|---|---|---|
| **BrainLat (BioSemi 128)** | 128 | 35 | 19 | 30 | 44 |
| **ds004504 (10–20, 19 ch)** | 19 | 7 | 1 | 4 | 7 |

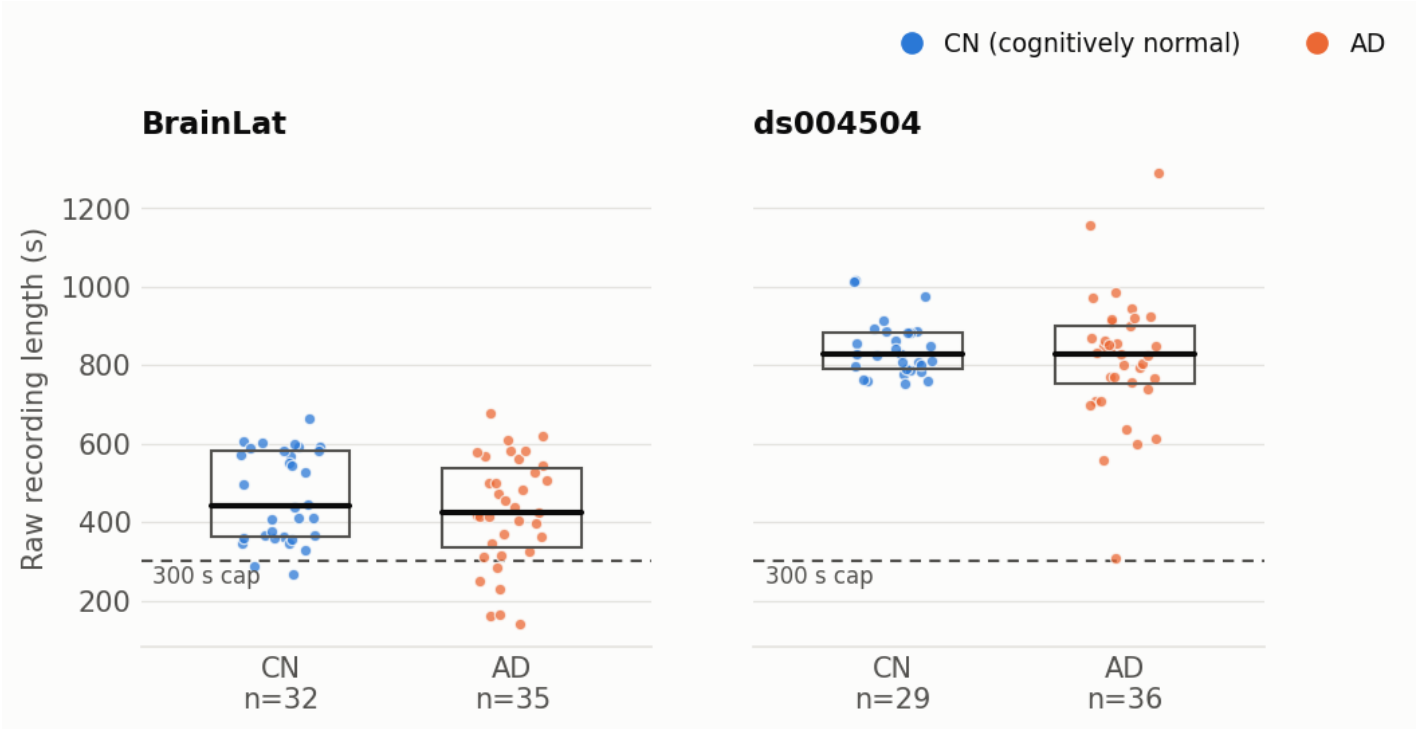



Figure 5. Raw recording length before the 300 s cap. 6 BrainLat AD recordings are shorter than 300 s (minimum 141 s), so those subjects contribute less data.

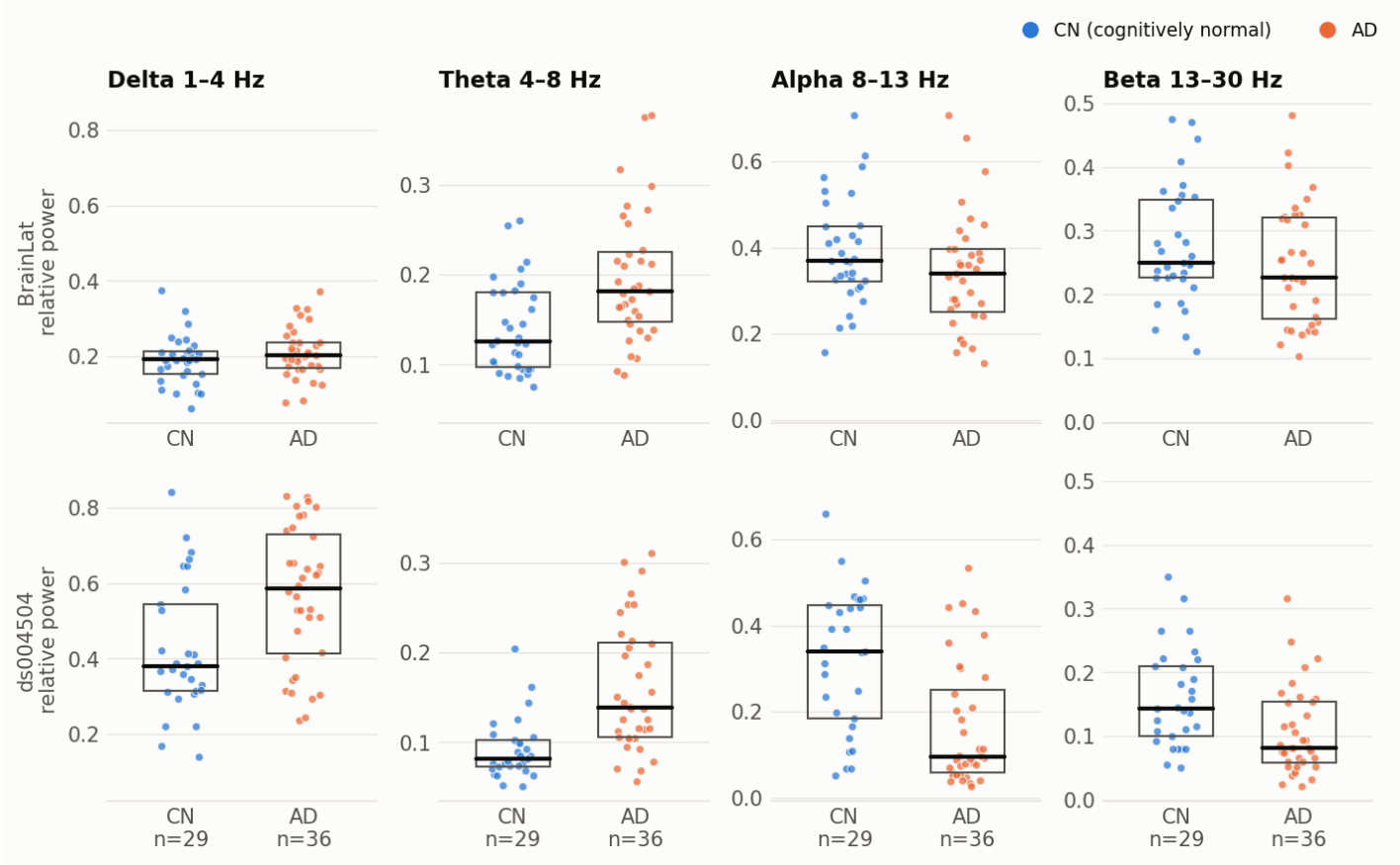



Figure 6. Global relative band power (fraction of 1–30 Hz power). Top row BrainLat, bottom row ds004504; each column shares its y-axis.

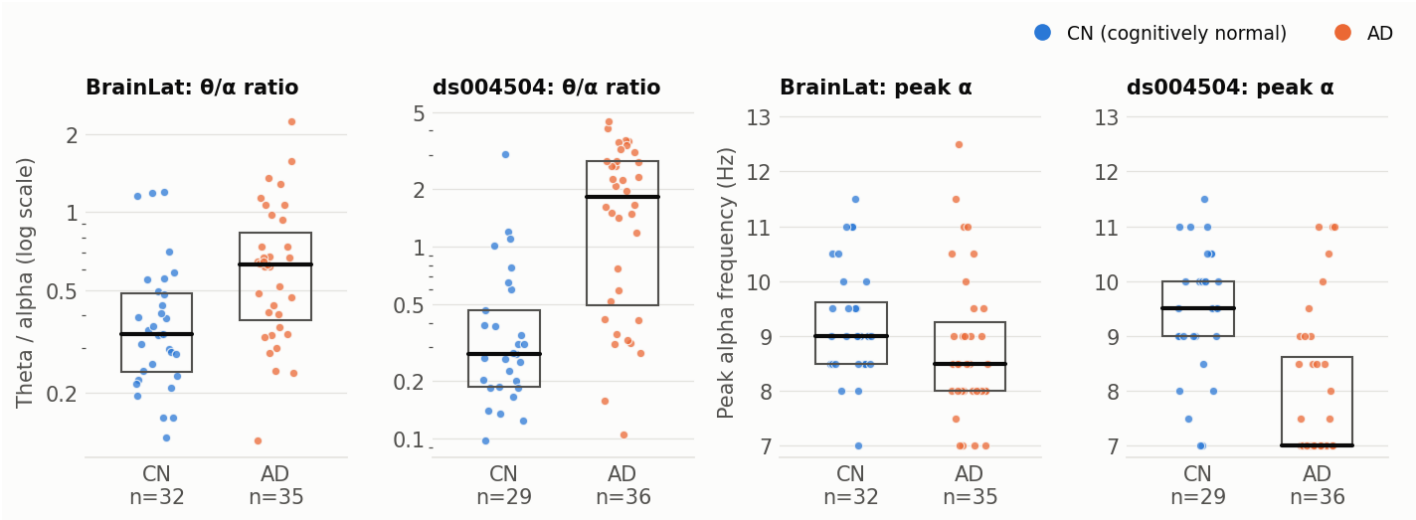



Figure 7. Theta/alpha ratio (log scale) and peak alpha frequency. AD shows EEG slowing in both datasets: higher θ/α and lower peak alpha. Peak alpha has 0.5 Hz resolution (2 s Welch windows) and is searched in 7–13 Hz, so values at 7 Hz (the ds004504 AD median) mean the spectral peak is at or below the search floor.

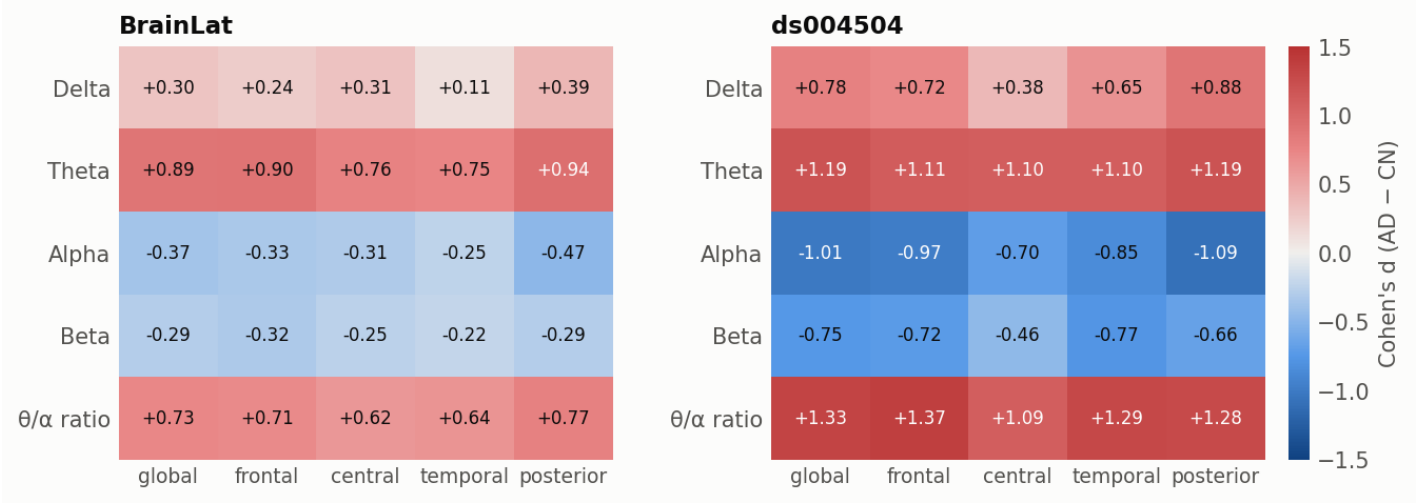



Figure 8. Effect size (Cohen's d, AD – CN) for each band and region. Red = higher in AD, blue = lower in AD.

## Global EEG features: AD vs CN

| Dataset | Feature | AD median | CN median | Cohen's d (AD–CN) | Mann–Whitney p |
|---|---|---|---|---|---|
| BrainLat | relpow_delta_global | 0.20 | 0.19 | 0.30 | 0.226 |
| BrainLat | relpow_theta_global | 0.18 | 0.13 | 0.89 | <0.001 |
| BrainLat | relpow_alpha_global | 0.34 | 0.37 | -0.37 | 0.118 |
| BrainLat | relpow_beta_global | 0.23 | 0.25 | -0.29 | 0.198 |
| BrainLat | theta_alpha_ratio_global | 0.63 | 0.34 | 0.73 | 0.001 |
| BrainLat | peak_alpha_freq | 8.50 | 9.00 | -0.38 | 0.029 |
| ds004504 | relpow_delta_global | 0.59 | 0.38 | 0.78 | 0.009 |
| ds004504 | relpow_theta_global | 0.14 | 0.08 | 1.19 | <0.001 |
| ds004504 | relpow_alpha_global | 0.09 | 0.34 | -1.01 | <0.001 |
| ds004504 | relpow_beta_global | 0.08 | 0.14 | -0.75 | 0.003 |
| ds004504 | theta_alpha_ratio_global | 1.80 | 0.28 | 1.33 | <0.001 |
| ds004504 | peak_alpha_freq | 7.00 | 9.50 | -1.16 | <0.001 |

## 5. Confounds and caveats

| Issue | Dataset | Details | What to do |
|---|---|---|---|
| **Recording type = diagnosis** | BrainLat | All AD files are `rs-HEP`; all CN files are `rs`. Any AD/CN difference may partly reflect the recording protocol. | Confirm that rs-HEP is the same eyes-closed rest protocol; otherwise treat BrainLat results as exploratory and use ds004504 for replication. |
| **Education** | BrainLat, ADNI | BrainLat AD 9.3 vs CN 15.6 years; ADNI 15.5 vs 16.6. Not recorded in ds004504. | Covariate where available. |
| **Age** | BrainLat | AD 76.5 vs CN 69.8 years. Matched in ADNI and roughly similar in ds004504. | Covariate, or age-matched subset. |
| **Sex** | ds004504 | AD 67% female vs CN 38%. | Covariate. |
| **Recording length** | BrainLat | 6 AD recordings < 300 s (as short as 141 s); 2 CN < 300 s. | Sensitivity analysis at a common shorter cap (e.g. 140 s). |
| **MMSE ceiling** | ds004504 | Every CN subject scores 30; MMSE is only available here. | Use only as an AD severity measure within AD. |
| **Montage density** | EEG, across datasets | 128 vs 19 channels; region definitions are the same, but the electrodes behind them are not. | Prefer global features for cross-dataset comparison. |
| **ADNI phase / QC** | ADNI | AD:CN ratio varies by phase; 769 of 870 FreeSurfer runs are not QC-rated. | Phase as covariate; check that results hold in the QC-rated subset. |
| **Not the same subjects** | All | EEG and MRI come from different cohorts (no subject has both). | Cross-modal analyses compare group-level effects, not paired data. |

## 6. Files

| File | Contents |
|---|---|
| `processed/mri/adni_mri_cohort.csv` | One row per ADNI subject: IDs, dx, scan info, demographics, 324 FreeSurfer measures |
| `processed/eeg_full/eeg_features.csv` | BrainLat AD/CN EEG features (one row per subject) |
| `processed/eeg_full/eeg_inventory.csv`, `eeg_missing_subjects.csv` | Status of every BrainLat .set file; subjects without usable data |
| `processed/ds004504/eeg_features.csv` | ds004504 AD/CN EEG features, same columns as BrainLat plus `mmse` |
| `processed/eeg/brainlat_verification.csv`, `processed/ds004504_verification.csv` | Download checks (file present, size, loads, sfreq, channels, duration) |
| `processed/eeg/synapse_ad_search.csv` | Every AD file on BrainLat Synapse (metadata) |

## Acknowledgments and data sources

Data collection and sharing for this project was funded by the Alzheimer's Disease Neuroimaging Initiative (ADNI) (National Institutes of Health Grant U01 AG024904) and DOD ADNI (Department of Defense award number W81XWH-12-2-0012). Data used in preparation of this work were obtained from the ADNI database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

BrainLat EEG: Prado P. et al., The BrainLat project, a multimodal neuroimaging dataset of neurodegeneration from underrepresented backgrounds, Scientific Data 10, 889 (2023); Synapse syn51549340.

ds004504 EEG: Miltiadous A. et al., A dataset of scalp EEG recordings of Alzheimer's disease, frontotemporal dementia and healthy subjects from routine EEG, Data 8(6), 95 (2023); OpenNeuro ds004504 (CC0).